

# CSE 4512

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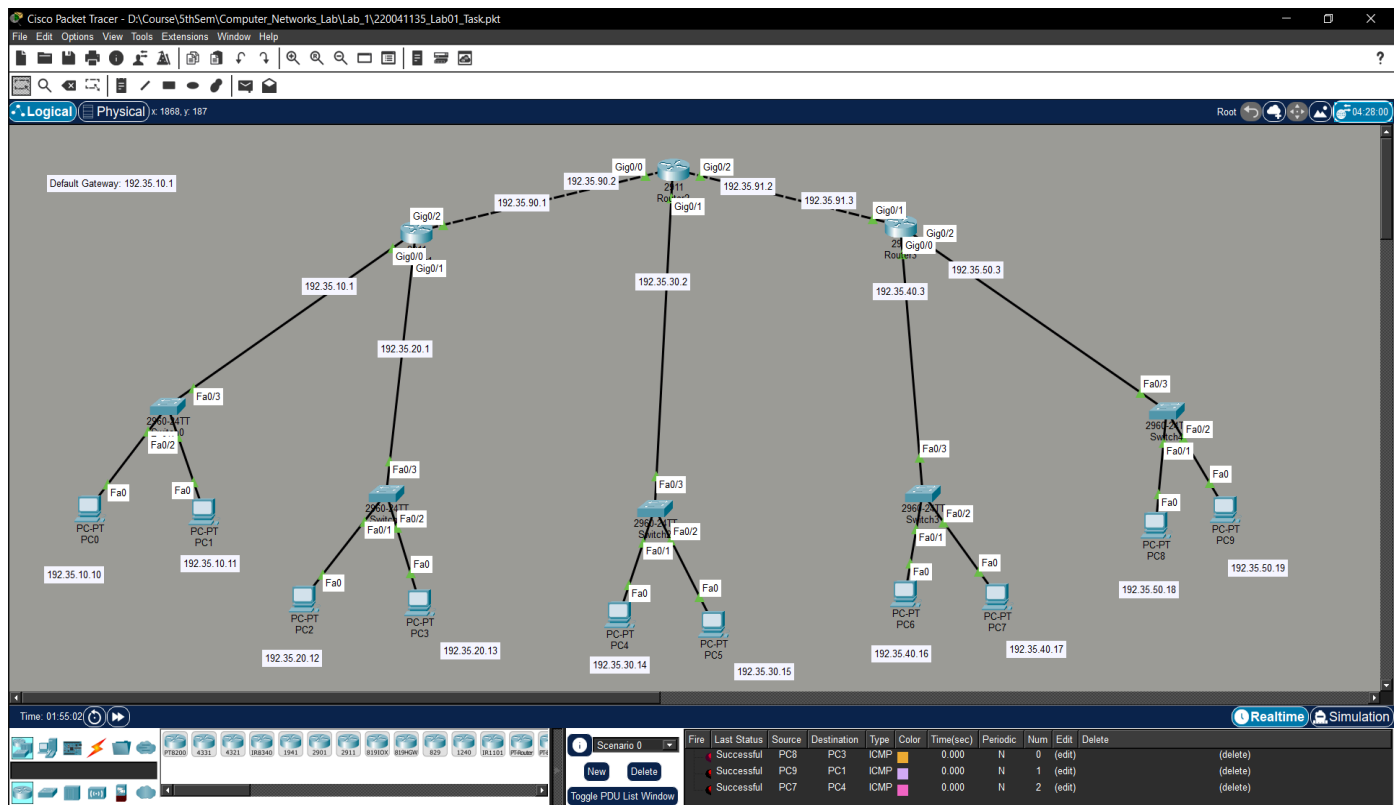
## Lab 01 Task

In Islamic University of Technology (IUT), each computer lab has its own network. There are five labs, each with two PCs. The lab's network has a pattern of **192.id** for the first 2 portions of the network address. **id** is the last two digits of your student id

Your task is to design an interconnection such that any PC in Lab x can reach any PC in Lab y.

As IUT is in Board Bazar, there is a problem of frequent load shedding. For that, sometimes some of the network connecting devices might not work right. Keep this in mind when designing the network to solve this issue.

## Final Topology



## Procedure

1. Firstly, as per the instructions two PCs were connected under one switch within the same network for a single lab and every lab has its own network (Different IP address).
2. 3 routers were installed as they contained only 3 ports to sufficiently supply ethernet connections to all the labs with the limited number of ports.
3. This also prevents the shutdown of whole networks in case of any router or device failure.
4. We gave ethernet gig 0/0 of router 1 as the default gateway for every PCs i.e 192.35.10.1
5. We then adjusted the gigabit ethernet interface using CLI to have the same network address as their respective lab for the 3 routers.
6. Interconnection between the routers were established between router1 and router2 & router2 with router 3 using a separate ip address.
7. After setting up all the IP addresses I have 'up' the gigabit ethernet using 'no shutdown' in CLI.
8. Thus the connection was established but we still needed routing.
9. Finally I established routing in every router using CLI so that they may know the ip address of other networks in order to send messages to the PCs which are not within the same network bound.
10. I have added IP addresses as a note in the topology ss for your convenience.

### Router 1

```
enable
config terminal
router rip
version 2
network 192.35.10.0
network 192.35.20.0
network 192.35.90.0
end
```

### Router 2

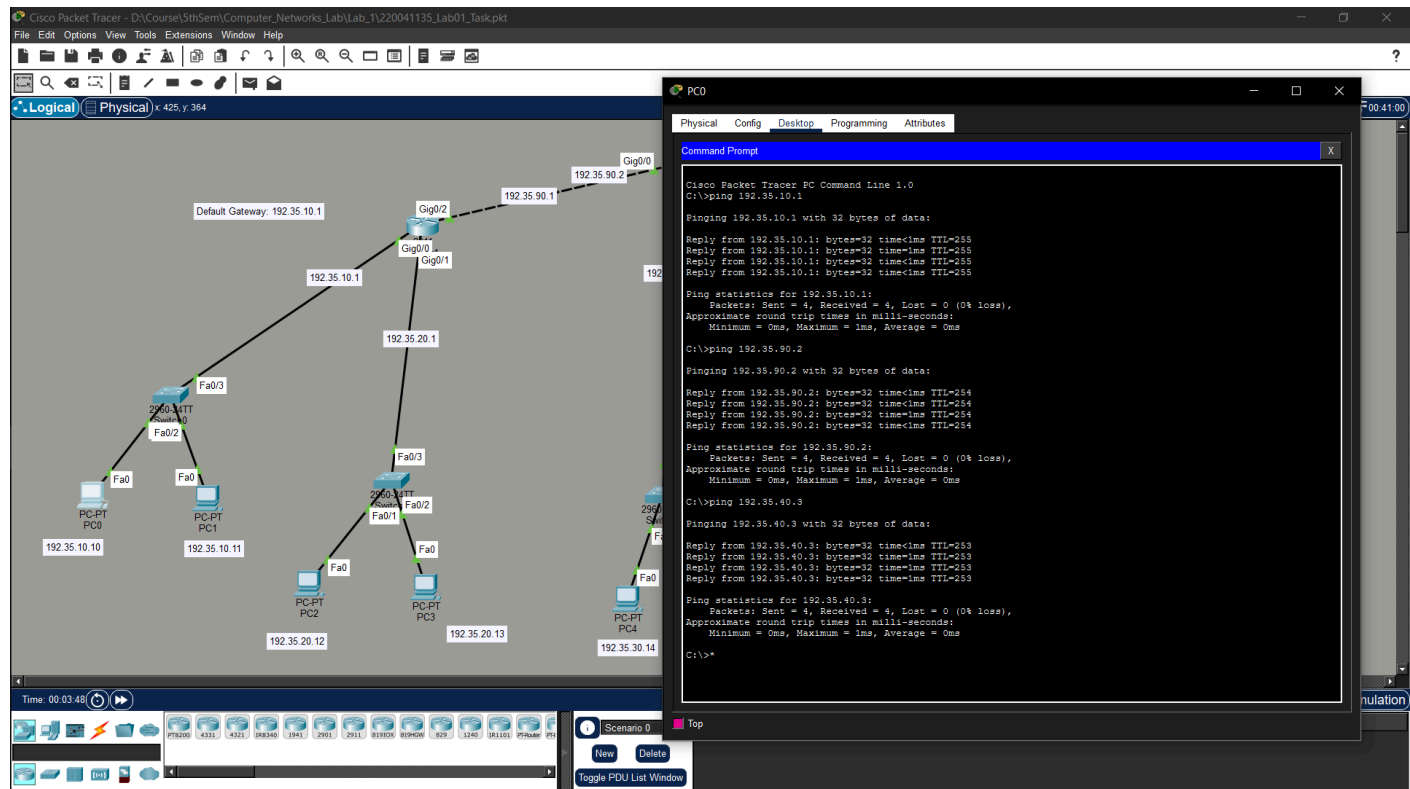
```
enable
config terminal
router rip
version 2
network 192.35.90.0
network 192.35.30.0
network 192.35.91.0
end
```

### Router 3

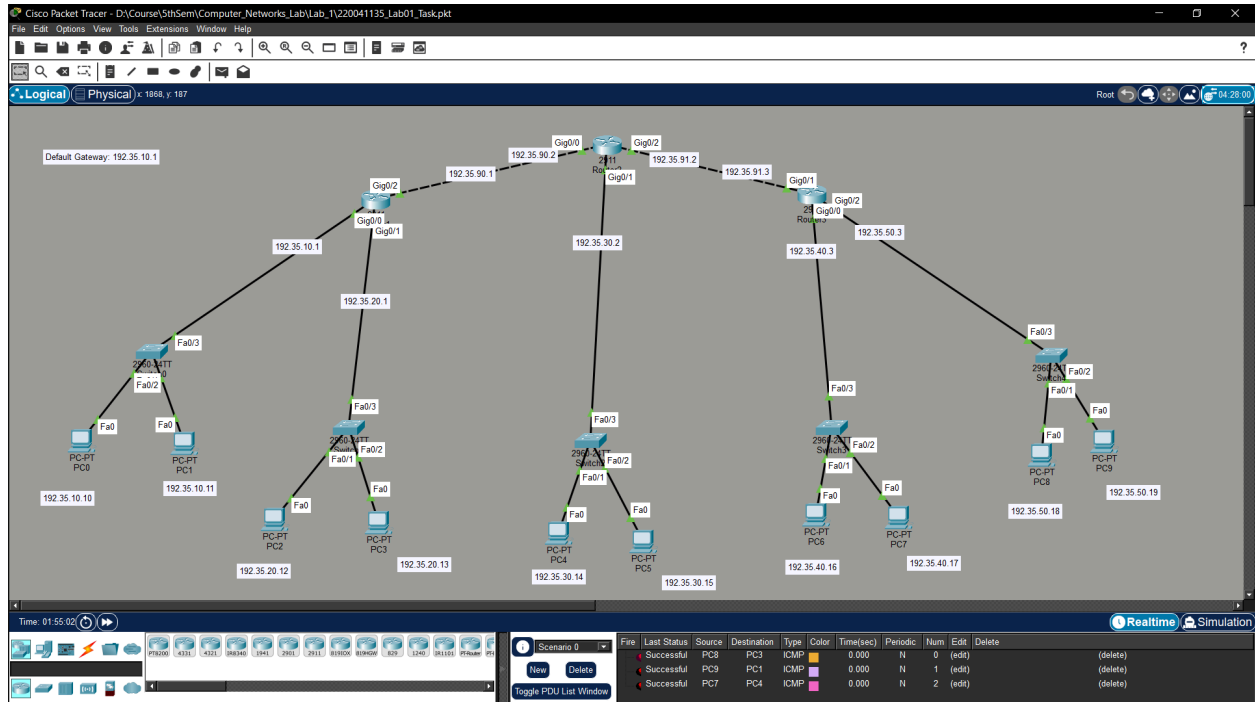
```
enable
config terminal
router rip
version 2
network 192.35.40.0
network 192.35.50.0
network 192.35.91.0
end
```

## Observations & Results

- Ping Checking for Lab1 PCs
  - Here in the following ss, we can see that from PC 0 we can send packets to every router without any errors

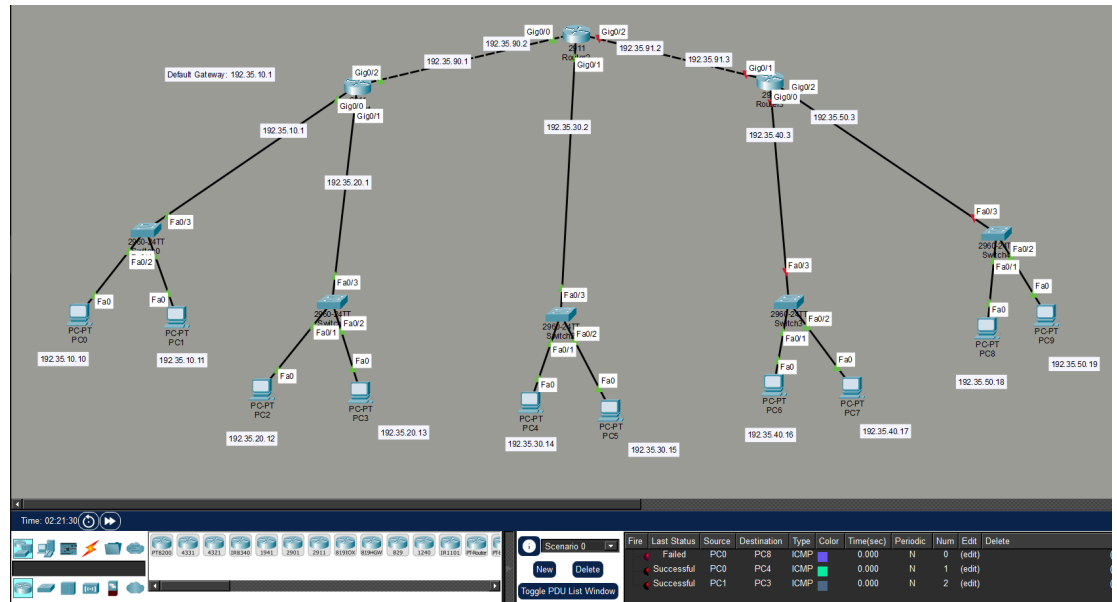


- Sending messages
  - In the following SS in the bottom right you can clearly see messages were sent successfully from PC8 ->PC3, PC9->PC1, PC7->PC4. Thus my network is fully up and running and each of the devices can communicate with each other.

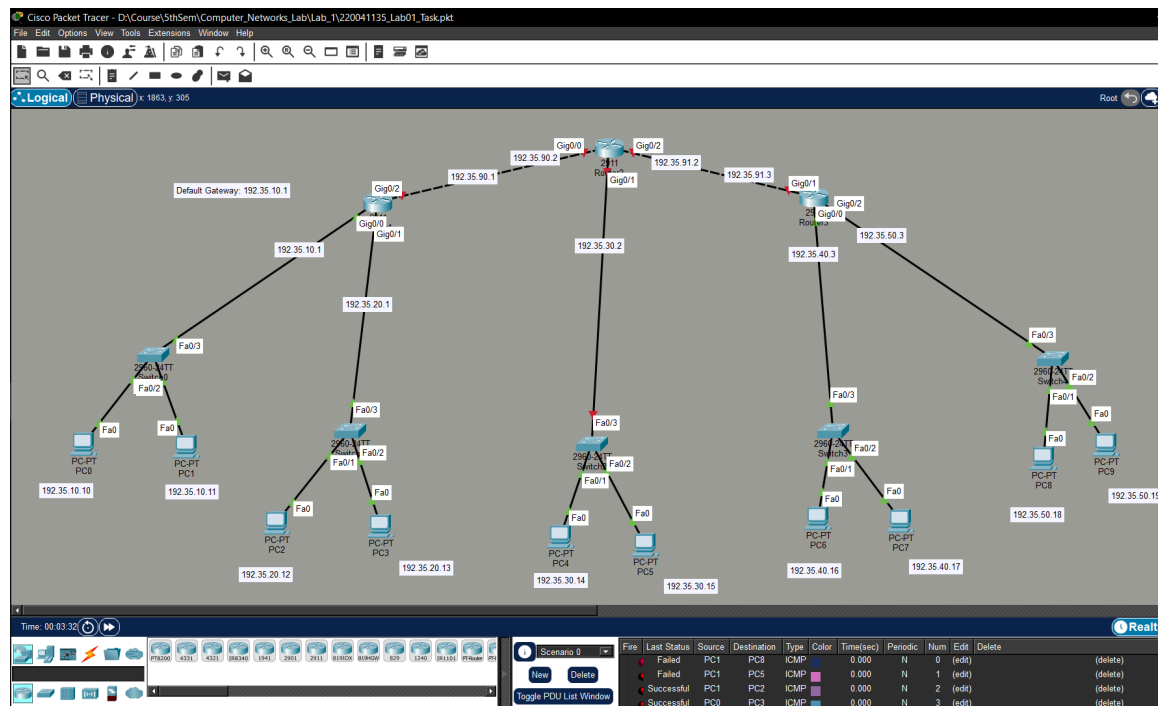


- Shutdown or Fault Handler

- Case-1: PC Shutdown:
  - In case of a load shedding or for any other reason if a PC shuts down the whole network will still be alive and running.
- Case-2: Switch Shutdown:
  - In case of switch shutdown the lab connected to that switch won't work. Other labs will still be running.
- Case-3: Router shutdown:
  - If Router 3 shuts down we can still communicate with the labs connected with router 1 and router 2. (shown in the following ss). As you can see PC0 cant communicate with PC8 but the messages is successful for PC0->PC4 and for PC1->PC3

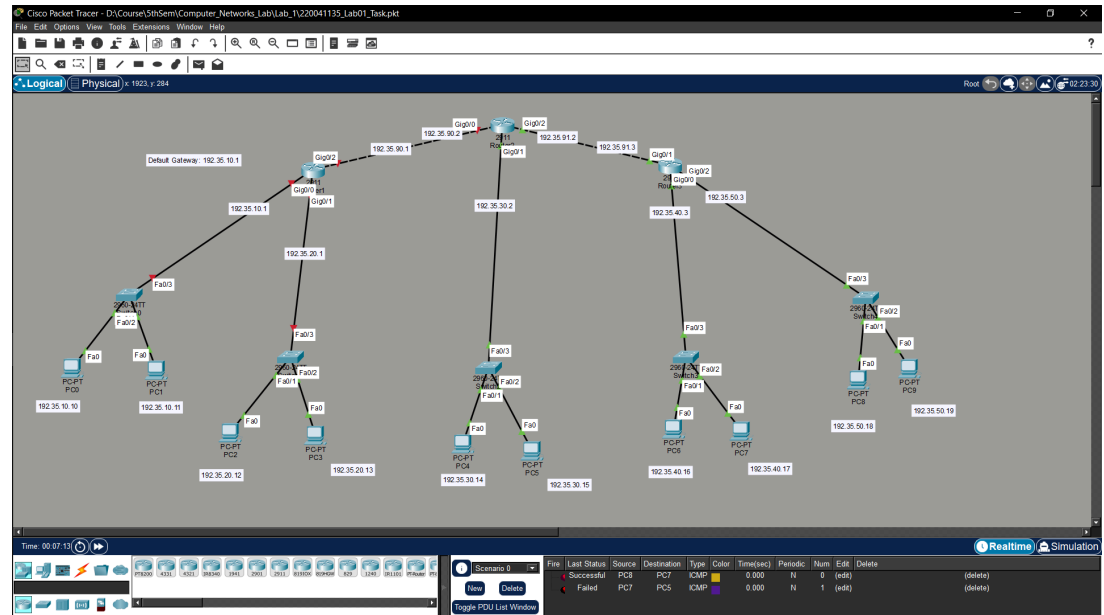


- What if Router 2 shuts down. Then the lab connected to my router 3 as well as router 2 doesn't work (My design limitations). I couldn't think of a way to use the three ports to connect between router 1 and router 3. As the ports are already occupied.



- Case-4: Router with default gateway shutdown:

- As all PCs have router 1 ethernet gig0/0 as the default gateway if router 1 fails the PCs cannot communicate with other networks but can communicate with the same network since if a PC wants to send data to a device in a different network/subnet, it sends it to the default gateway. The router then forwards the traffic toward the destination network.



### ● Possibilities or Improvements:

- There should have been mesh connections between routers (router1->router3) to handle cases like middle router failure to prevent network shutdown. But my ethernet ports are already exhausted. I don't know if there is any possibility of any serial external connection to extend ports in routers.
- Another possibility is to use another router in the bottom to connect the most important PCs which may be used in cases like a backup plan to prevent the communication breakages between the most important computers but to use 4 routers for 5 labs; i wonder if it is okay or not.
- Redundant routers might be used like another 3 routers at the bottom like the above 3 routers for backing up all the computers shutdown in case all the 3 routers at the above fails. Again 6 routers for 5 labs; I wonder if it's okay or not.

### Challenges

- Routing at the end was confusing as I cannot know whether the routing(rip) was established or not.

- Learning and implementing the interfaces and CLI as a beginner was a bit challenging
- Debugging is a bit tough
- Managing proper IP addresses throughout different networks is challenging. One faulty address and you are doomed.